Unit: Unit 7: Factoring
Topic: Sum and Difference of Cubes

Name: ________________
Date: ____________

Advanced (Habanero)

Q1. What is the factored form of $x^3 + 8$?
- (A) $(x+2)(x^2-2x+4)$
- (B) $(x-2)(x^2+2x+4)$
- (C) $(x+2)(x^2+2x+4)$
- (D) $(x-2)(x^2-2x+4)$
- (E) $(x+2)(x^2-2x-4)$

Q2. Factor $27x^3 - 64$
- (A) $(3x-4)(9x^2+12x+16)$
- (B) $(3x+4)(9x^2-12x+16)$
- (C) $(3x+4)(9x^2+12x+16)$
- (D) $(3x-4)(9x^2-12x+16)$
- (E) $(3x+4)(9x^2-12x-16)$

Q3. What is the factored form of $x^3 - 64y^3$?
- (A) $(x-4y)(x^2+4xy+16y^2)$
- (B) $(x+4y)(x^2-4xy+16y^2)$
- (C) $(x-4y)(x^2-4xy+16y^2)$
- (D) $(x+4y)(x^2+4xy+16y^2)$
- (E) $(x-4y)(x^2+4xy-16y^2)$

Q4. Factor $8x^3 + 27y^3$
- (A) $(2x+3y)(4x^2-6xy+9y^2)$
- (B) $(2x-3y)(4x^2+6xy+9y^2)$
- (C) $(2x+3y)(4x^2+6xy+9y^2)$
- (D) $(2x-3y)(4x^2-6xy+9y^2)$
- (E) $(2x+3y)(4x^2-6xy-9y^2)$

Q5. What is the factored form of $x^3 - 1000$?
- (A) $(x-10)(x^2+10x+100)$
- (B) $(x+10)(x^2-10x+100)$
- (C) $(x+10)(x^2+10x+100)$
- (D) $(x-10)(x^2-10x+100)$
- (E) $(x+10)(x^2-10x-100)$

Q6. Factor $125x^3 - 8y^3$
- (A) $(5x-2y)(25x^2+10xy+4y^2)$
- (B) $(5x+2y)(25x^2-10xy+4y^2)$
- (C) $(5x-2y)(25x^2-10xy+4y^2)$
- (D) $(5x+2y)(25x^2+10xy+4y^2)$
- (E) $(5x-2y)(25x^2+10xy-4y^2)$

Q7. What is the factored form of $8x^3 + 125y^3$?
- (A) $(2x+5y)(4x^2-10xy+25y^2)$
- (B) $(2x-5y)(4x^2+10xy+25y^2)$
- (C) $(2x+5y)(4x^2+10xy+25y^2)$
- (D) $(2x-5y)(4x^2-10xy+25y^2)$
- (E) $(2x+5y)(4x^2-10xy-25y^2)$

Q8. Factor $64x^3 - 27y^3$
- (A) $(4x-3y)(16x^2+12xy+9y^2)$
- (B) $(4x+3y)(16x^2-12xy+9y^2)$
- (C) $(4x-3y)(16x^2-12xy+9y^2)$
- (D) $(4x+3y)(16x^2+12xy+9y^2)$
- (E) $(4x-3y)(16x^2+12xy-9y^2)$

Open Ended Questions (FRQ)

Q9. Factor $x^3 + 64y^3$ completely

Q10. Factor $27x^3 - 8y^3$ completely

Chef's Tips

- Check for common factors
- Use sum and difference of cubes formulas
- Simplify expressions completely